Dorey's Resource Book

Topic 2 – Relative frequency and probability

Contents

Episode 1. The probability scale

1 The probability scale

ANCHOR QUESTION

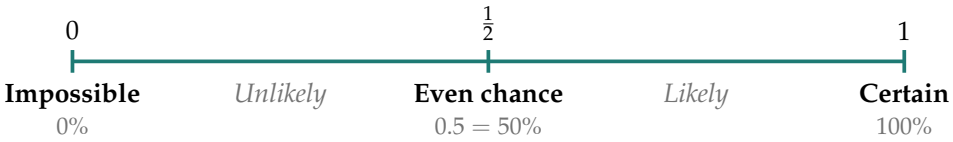
The probability that it rains tomorrow is 0.8. *Write this as a percentage and describe the chance in words.*

SYLLABUS MST-12-S1-07

- Recognise that probabilities range from 0 (impossible) to 1 (certain) and can be represented as fractions, decimals and percentages

NOTES

- **Probability** measures how likely an event is to happen.
- Every probability is between 0 and 1, including 0 and 1. As a percentage, it is between 0% and 100%.
- 0 means **impossible**, $\frac{1}{2}$ means **even chance** and 1 means **certain**.
- Between 0 and $\frac{1}{2}$ is **unlikely**. Between $\frac{1}{2}$ and 1 is **likely**.
- A probability can be written as a fraction, a decimal or a percentage: $\frac{1}{4} = 0.25 = 25\%$.
- To compare probabilities, write them all in the same form first.



WORKED EXAMPLE

The probability that a randomly chosen phone battery lasts more than two days is 0.25. *Write this as a percentage and describe the chance in words.*

$0.25 \times 100 = 25\%$ — decimal to percentage: $\times 100$
25% is less than 50% — compare with even chance
The chance is **unlikely**. — between 0 and $\frac{1}{2}$

Tier 1 · Decimal to percentage and words

1. Which of these numbers could be a probability?

$$0.4 \qquad 1.3 \qquad 75\% \qquad -0.1 \qquad 120\% \qquad \frac{2}{3}$$

2. Describe each probability in words.
(a) 0 (b) $\frac{1}{2}$ (c) 0.7 (d) 100%

3. The probability that a train arrives on time is 0.9. Write this as a percentage and describe the chance in words.

4. The probability that a new light globe is faulty is 0.02. Write this as a percentage and describe the chance in words.

Tier 2 · Starting from a fraction

5. The probability that a randomly chosen student walks to school is $\frac{2}{5}$. Write this as a decimal and as a percentage, and describe the chance in words.

6. The probability that a flight leaves on time is $\frac{17}{20}$. Write this as a decimal and as a percentage, and describe the chance in words.

7. The probability of winning a prize in a game is $\frac{1}{8}$. Write this as a decimal and as a percentage, and describe the chance in words.

Tier 3 · Starting from a percentage

8. A forecast gives a 60% chance of rain. Write this as a fraction in simplest form and as a decimal, and describe the chance in words.

9. In a raffle, 4% of the tickets win a prize. Write the probability that a ticket wins as a fraction in simplest form and as a decimal, and describe the chance in words.

10. A basketballer makes 72% of her free throws. Write the probability that she makes her next free throw as a fraction in simplest form and as a decimal, and describe the chance in words.

Tier 4 · Comparing and ordering

11. Order these probabilities from least likely to most likely: 0.3, $\frac{2}{5}$, 35%.

12. Order these probabilities from least likely to most likely: $\frac{7}{8}$, 0.85, 88%.

13. The probabilities of three teams winning a competition are: Hawks 0.45, Lions $\frac{9}{20}$, Owls 44%. Which team is least likely to win? Are any two teams equally likely to win?

Tier 5 · Judging from the situation

14. Use one of the words *impossible*, *unlikely*, *even chance*, *likely* or *certain* to describe each event.

(a) A fair coin lands on heads.

(b) A standard six-sided die is rolled and shows a 7.

(c) A randomly chosen month has at least 28 days.

15. One student from a class of 30 is chosen at random to be class captain. Which of the following best describes the probability that Ella, a student in the class, is chosen?

A. Certain

B. Even chance

C. Unlikely

D. Impossible

16. A bag holds 9 green counters and 1 red counter. A counter is taken out without looking. Describe, in words, the chance that it is green.

Conceptual

17. Explain why 1.4 cannot be a probability.

18. Sam says a probability of 0.08 is more likely than a probability of 0.5 because 8 is bigger than 5. Explain his mistake.

19. An event has an even chance of happening. Write its probability as a fraction, a decimal and a percentage.

20. A weather app gives a 0% chance of rain for tomorrow. Explain what this means, and why it might still rain.

21. The probability that a phone made at Factory A is faulty is 0.03.

(a) Write this probability as a percentage. (1 mark)

(b) Describe this chance in words. (1 mark)

(c) The probability that a phone made at Factory B is faulty is $\frac{1}{25}$. Which factory is more likely to make a faulty phone? Justify your answer. (2 marks)

Revision

22. A fruit punch is made by mixing juice and soda water in the ratio 3 : 2.

(a) What fraction of the punch is soda water? (1 mark)

(b) How many litres of juice are needed to make 4 L of punch? (2 marks)

(c) Juice costs \$3.60 per 2 L bottle. Find the cost of the juice for 4 L of punch. (2 marks)

23. Kira drives 342 km in 3 hours 48 minutes.

(a) Write 3 hours 48 minutes in hours, as a decimal. (1 mark)

(b) Find Kira's average speed in km/h. (1 mark)

(c) Convert this speed to m/s. (1 mark)

(d) Her car uses 7.5 L/100 km and petrol costs \$1.96 per litre. Find the cost of the fuel for the trip, to the nearest cent. (2 marks)

24. The number of goals scored by a netball team in eight games is shown.

4 7 7 8 9 10 12 15

(a) Find the mean. (1 mark)

(b) Find the median. (1 mark)

(c) Find the range. (1 mark)

(d) Find the interquartile range. (2 marks)

Solutions

1. 0.4, 75% and $\frac{2}{3}$. The others are less than 0 or more than 1.

2. (a) Impossible. (b) Even chance. (c) Likely. (d) Certain.

3. $0.9 \times 100 = 90\%$. Likely.

4. $0.02 \times 100 = 2\%$. Unlikely.

5. $2 \div 5 = 0.4 = 40\%$. Unlikely.

6. $17 \div 20 = 0.85 = 85\%$. Likely.

7. $1 \div 8 = 0.125 = 12.5\%$. Unlikely.

8. $\frac{60}{100} = \frac{3}{5}$, 0.6. Likely.

9. $\frac{4}{100} = \frac{1}{25}$, 0.04. Unlikely.

10. $\frac{72}{100} = \frac{18}{25}$, 0.72. Likely.

11. 0.3, 0.35, 0.4, so 0.3, 35%, $\frac{2}{5}$.

12. $\frac{7}{8} = 0.875$. In order: 0.85, $\frac{7}{8}$, 88%.

13. Lions: $9 \div 20 = 0.45$. Owls: 0.44. The Owls are least likely. The Hawks and Lions are equally likely (0.45 each).

14. (a) Even chance. (b) Impossible. (c) Certain.

15. C. Ella is 1 of 30 students, so it could happen but the chance is small.

16. Likely. Most of the counters (9 out of 10) are green.

17. A probability is between 0 and 1. 1.4 is more than 1, which would be more likely than certain.

18. $0.08 = 8\%$ and $0.5 = 50\%$. Since $8\% < 50\%$, 0.5 is more likely. He compared the digits, not the values.

19. $\frac{1}{2} = 0.5 = 50\%$

20. 0% means rain is impossible. A forecast is only an estimate, often rounded, so rain can still happen.

21. (a) $0.03 \times 100 = 3\%$. (b) Unlikely. (c) $\frac{1}{25} = 0.04 = 4\%$. Since $4\% > 3\%$, Factory B is more likely to make a faulty phone.

22. (a) $3 + 2 = 5$ parts, so $\frac{2}{5}$. (b) One part $= 4 \div 5 = 0.8$ L. Juice $= 3 \times 0.8 = 2.4$ L. (c) $\$3.60 \div 2 = \$1.80$ per litre. $2.4 \times \$1.80 = \$4.32$.

23. (a) $48 \div 60 = 0.8$, so 3.8 h. (b) $342 \div 3.8 = 90$ km/h. (c) $90\,000 \div 3600 = 25$ m/s. (d) $7.5 \times 3.42 = 25.65$ L. $25.65 \times \$1.96 = \$50.274 \approx \$50.27$.

24. (a) $72 \div 8 = 9$. (b) $\frac{8+9}{2} = 8.5$. (c) $15 - 4 = 11$. (d) Q_1 is the median of $4, 7, 7, 8$, so $Q_1 = 7$. Q_3 is the median of $9, 10, 12, 15$, so $Q_3 = 11$. IQR $= 11 - 7 = 4$.